

Original software publication

# M3ALEM+: A smart geolocated service marketplace with real-time matching, secure payment, and AI-based fraud detection

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| ARTICLE INFO                                                                                                                                                                            | ABSTRACT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Keywords:</b><br>Geolocated marketplace<br>Real-time matching<br>Mobile application<br>Fraud detection<br>Secure payment<br>Service platform<br>Recommender systems<br>Microservices | The informal economy for skilled artisans remains fragmented, making it difficult for clients to locate qualified providers. This paper presents M3ALEM+, a full-stack mobile platform for geolocation-based matching between service seekers and local artisans. The platform integrates real-time request dispatching with quality-driven ranking, bidirectional WebSocket chat, AI-assisted recommendation and fraud detection, and secure escrow payment. The backend uses Django REST Framework with Django Channels, while the mobile frontend uses React Native (Expo). An AI microservice handles recommendation and anomaly detection for fraudulent activities. The system is containerised with Docker and deployable as a production-grade application. |
| <b>Code metadata</b>                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Current code version                                                                                                                                                                    | V1.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Permanent link to code/repository                                                                                                                                                       | <a href="https://github.com/achraf-bouchra/m3alem-plus">https://github.com/achraf-bouchra/m3alem-plus</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Permanent link to reproducible capsule                                                                                                                                                  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Code versioning system used                                                                                                                                                             | git                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Software code languages, tools, and services used                                                                                                                                       | Python 3.11, JavaScript (React Native), Django 4.2, Django REST Framework, Django Channels 4.0, MySQL 8.0, Redis 7, scikit-learn 1.3, Docker 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Compilation requirements, operating environments and dependencies                                                                                                                       | Docker Compose 2.20+; Python >= 3.10; Node.js >= 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| If available, link to developer documentation/manual                                                                                                                                    | See README.md in repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Support email for questions                                                                                                                                                             | <a href="mailto:achrafadardor345@gmail.com">achrafadardor345@gmail.com</a> , <a href="mailto:bouchraouaqa@gmail.com">bouchraouaqa@gmail.com</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Software metadata</b>                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Current software version                                                                                                                                                                | V1.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Permanent link to executables of this version                                                                                                                                           | N/A (source code + Docker deployment)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Legal Software License                                                                                                                                                                  | MIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Computing platforms / Operating Systems                                                                                                                                                 | Android 10+ (React Native); Linux Mac, windows 22.04 LTS (backend)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Installation requirements & dependencies                                                                                                                                                | Docker >= 24.0; Docker Compose >= 2.20; Android Studio 2023+ or Xcode 15+                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| If available, link to user manual                                                                                                                                                       | <a href="https://github.com/achraf-bouchra/m3alem-plus/blob/main/README.md">https://github.com/achraf-bouchra/m3alem-plus/blob/main/README.md</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Support email for questions                                                                                                                                                             | <a href="mailto:achrafadardor345@gmail.com">achrafadardor345@gmail.com</a> , <a href="mailto:bouchraouaqa@gmail.com">bouchraouaqa@gmail.com</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## 1. Motivation and significance

The on-demand local services market encompassing plumbers, electricians, carpenters, painters, and similar skilled tradespeople suffers from persistent informational asymmetries that reduce economic efficiency and trust. Clients typically rely on word-of-mouth referrals, unstructured social media queries, or fragmented web directories to locate available artisans, while service providers lack efficient digital channels to reach nearby customers in real time [1,2]. These frictions increase response latency, reduce market efficiency, and expose both parties to trust and payment risks. In developing and emerging-market contexts, particularly across the Middle East and North Africa (MENA) region, these challenges are compounded by limited penetration of formal service platforms and a predominantly cash-based informal economy [3].

Digital platform solutions for skilled trades have emerged in various forms, ranging from static web directories to appointment-booking applications. However, the majority of existing solutions lack three capabilities that are critical for a seamless user experience in time-sensitive service contexts: (i) real-time proximity-based artisan discovery and instant dispatch mechanisms comparable to those employed in ride-hailing platforms [4]; (ii) bidirectional, low-latency in-app communication that eliminates the need for external messaging applications; and (iii) automated trust mechanisms that can flag fraudulent ratings, fake reviews, or anomalous platform behaviour before they degrade service quality [5,6].

M3ALEM+ addresses these gaps by proposing an integrated mobile platform that combines intelligent geolocation matching, a real-time request lifecycle management system, an AI-powered recommendation and fraud detection service, and a secure end-to-end payment workflow. The name is derived from the Arabic term *m3alem* meaning master craftsman or skilled professional, reflecting the platform's focus on the Maghrebi and broader MENA informal labour context, while remaining architecturally generalisable to other regional settings and service categories.

From a scientific software perspective, M3ALEM+ constitutes a reusable reference implementation demonstrating the integration of heterogeneous modern technologies including asynchronous Django, React Native, and lightweight machine learning microservices within a coherent, deployable system. The principal contributions presented in this paper are: (1) a real-time geolocation-based service matching engine with composite quality scoring; (2) a quality-ranked artisan recommendation model incorporating collaborative filtering; (3) a behavioural anomaly and fake-review detection module based on Isolation Forest; (4) a secure in-app payment integration with escrow-style fund holding; and (5) a containerised microservice architecture ready for production deployment and academic experimentation.

## 2. Software description

This section presents the M3ALEM+ platform through its layered architecture, component decomposition, and the principal functionalities accessible to end users.

### 2.1. System architecture

M3ALEM+ is developed as a layered microservice-based platform, as illustrated in Fig. 1. The system consists of a React Native mobile application built with Expo, enabling clients and artisans to access services through a unified cross-platform interface. The frontend communicates with a Django REST Framework backend via REST APIs for business operations and WebSocket connections for real-time messaging and notifications. A dedicated real-time service based on Django Channels and Redis supports instant communication and job-status updates. Recommendation and fraud detection functionalities are deployed as an independent AI microservice accessible through lightweight REST endpoints. Persistent data storage is provided by MySQL with spatial indexing support for efficient geolocation queries. The complete platform is containerised using Docker Compose to ensure reproducible deployment across different environments.

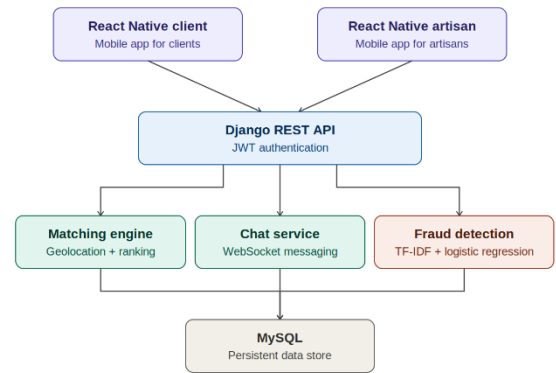


Fig. 1. System architecture of M3ALEM+

### 2.2. Software functionalities

Based on the proposed software architecture, the M3ALEM+ platform is designed and developed to support the complete lifecycle of home-service transactions between clients and artisans. The platform integrates geolocation-based artisan discovery, service request management, real-time communication, payment processing, and AI-assisted fraud detection functionalities. Before illustrating the operational workflow, several supporting mechanisms are implemented to facilitate platform operation, including user authentication and authorization, artisan profile verification, geospatial indexing, WebSocket-based communication, and machine-learning services for review moderation.

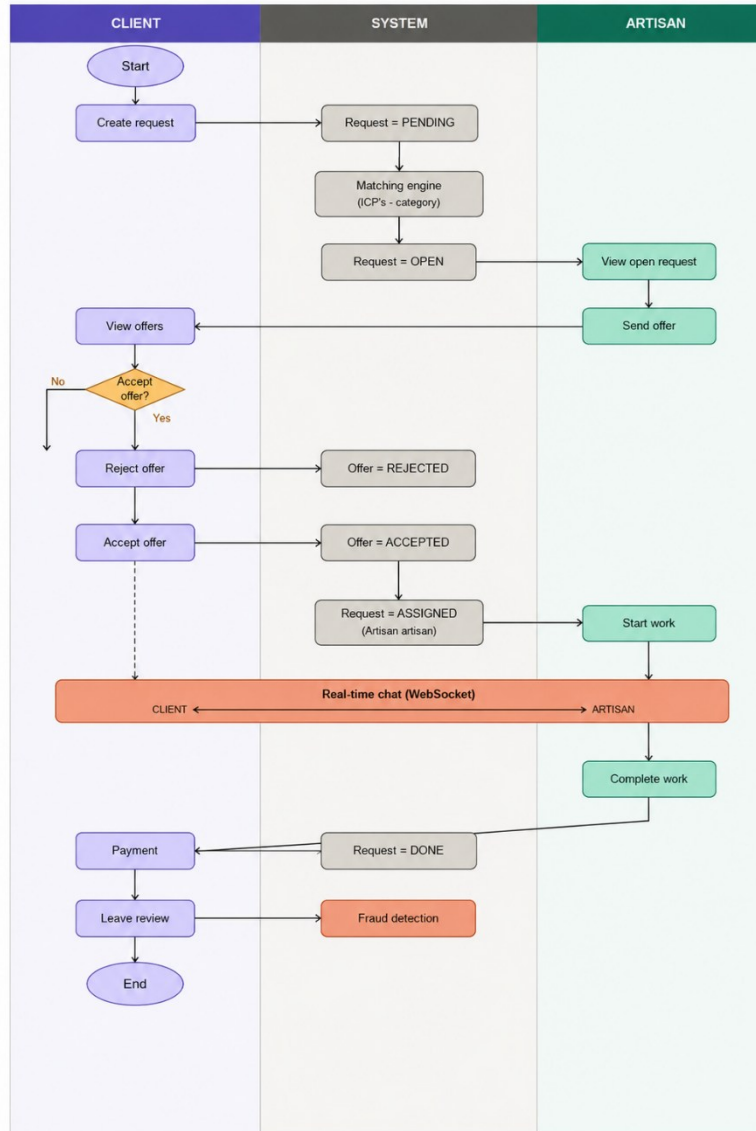


Fig. 2. M3ALEM+ platform workflow

When a client enters the platform, a service request can be created by selecting a service category, providing a textual description of the problem, and allowing the application to capture the current GPS coordinates of the device. The request information is transmitted to the backend server, where a spatial query is executed to identify artisans located within a configurable geographic radius and currently available to perform the requested service.

After retrieving the candidate artisans, the matching module applies a ranking mechanism to determine the most suitable service providers for the client request. For each candidate artisan, a matching score is calculated according to Eq. (1), where  $distance_{km}$  represents the geographical distance between the client and the artisan, and  $rating$  denotes the artisan's

average service rating on a five-point scale. The scoring function favours artisans who are geographically closer and highly rated by previous clients. Candidate artisans are subsequently sorted in ascending order of the computed score, such that lower scores correspond to higher matching priority.

$$Score = distance\_km + (5 - rating)$$

The resulting ranked list is then used to notify eligible artisans about the service request. Similar to on-demand service platforms, artisans may either respond by submitting a service offer or ignore the request. If no response is received within a predefined time interval, the request can be proposed to the next highest-ranked artisan until a suitable offer is obtained. This ranking mechanism enables efficient allocation of service requests while balancing geographical proximity and service quality considerations.

Upon receiving a service request, artisans may submit service offers containing the proposed service conditions and pricing information. The client is then allowed to evaluate the received offers and either accept or reject them. Once an offer is accepted, the selected artisan becomes assigned to the request, while all remaining offers associated with the same request are automatically rejected. This mechanism guarantees that only one artisan can be assigned to a service request at any given time.

Following offer acceptance, a dedicated communication channel is automatically established between the client and the artisan. The real-time communication module is implemented using Django Channels and WebSocket technology, allowing participants to exchange messages instantly throughout service execution. All messages are persisted in the database and synchronised through Redis-based channel layers. In addition to user-generated messages, the communication module distributes notifications related to service status updates and system events.

The service request itself follows a predefined lifecycle managed by the backend system. Initially, newly created requests are assigned the PENDING state. Once artisans become eligible to submit offers, the request enters the OPEN state. Following offer acceptance, the request transitions to the ASSIGNED state and remains active until service completion. After successful execution and payment confirmation, the request is marked as DONE. Additional states such as CANCELLED are supported to handle exceptional situations. All state transitions are recorded in the database to ensure traceability and support auditing procedures.

After the requested service has been completed, the payment workflow is initiated. The platform creates a payment transaction associated with the accepted offer, the client, and the assigned artisan. Transaction information includes the payment amount, selected payment method, transaction status, and fraud analysis results. Similar to the request lifecycle, payments progress through predefined states, namely PENDING, PAID, FAILED, REFUNDED, and FRAUD. Upon successful payment validation, the corresponding service request is automatically finalised and recorded as completed.

To improve platform trustworthiness, the M3ALEM+ platform incorporates an AI-assisted fraud detection module operating independently from the core application services. The fraud detection process primarily focuses on review authenticity analysis and suspicious behavioural patterns. When a client submits a rating or textual review, the content is transmitted to the AI microservice for analysis. The system computes a fraud probability score, a fake-review classification result, and a moderation status based on machine-learning predictions and rule-based validation mechanisms.

Reviews identified as potentially fraudulent are flagged for moderation and may be temporarily excluded from artisan reputation calculations until verification is completed. Fraud-related events are additionally stored as platform alerts to facilitate administrative investigations and maintain marketplace integrity. By combining automated machine-learning analysis with human moderation procedures, the

platform contributes to maintaining trust, fairness, and reliability within the digital marketplace ecosystem.

## 2.3. Dataset description

### 2.3.1. Dataset for fake review detection

The fraud detection component relies on a curated dataset of 71,885 user-generated reviews collected and preprocessed in 2023. The dataset covers local services and artisan activities in French and English, with four main features: category (service type), rating (1-5), label (0 = genuine, 1 = fake/spam), and text (review content).

| Property          | Value                                      |
|-------------------|--------------------------------------------|
| Number of reviews | 71,885 entries                             |
| Main features     | category, rating, label, text              |
| Target class      | CG/OR                                      |
| Domains           | Local services, artisans, small businesses |
| Language          | French / English                           |
| Collection year   | 2023                                       |

Table 1. Characteristics of the Fake Review

### 2.3.2. Analytical objectives

The dataset supports binary classification of reviews, detection of inconsistencies between textual sentiment and numeric ratings, and identification of suspicious patterns including repetitive content, coordinated rating bursts, and abnormal user activity profiles.

### 2.3.3. Integration within M3ALEM+

The dataset is integrated into the AI microservice, where scikit-learn-based models are trained to detect fraudulent reviews in real time. This mechanism enhances trust and ensures the

reliability of user-generated content within the platform ecosystem.

## 2.4. Performance evaluation of the fraud detection model

To evaluate the effectiveness of the proposed Isolation Forest-based fraud detection module, experiments were conducted on the held-out test split of the review dataset described in Section 3.3. The quantitative performance metrics are summarised in Table 3. The model achieved an accuracy of 91.79%, a precision of 93.02%, a recall of 90.33%, an F1-score of 91.65%, and a ROC-AUC of 97.28%, demonstrating its ability to accurately distinguish between genuine (OR) and fraudulent (CG) reviews while maintaining a good balance between precision and recall.

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 91.79% |
| Precision | 93.02% |
| Recall    | 90.33% |
| F1-score  | 91.65% |
| ROC-AUC   | 97.28% |

Table 3. Performance metrics of the fraud detection model

Figure 3 illustrates the confusion matrix of the proposed fraud detection model. Most original (OR) and computer-generated (CG) reviews are correctly classified, with relatively few misclassification errors. These results indicate that the model effectively balances false positives and false negatives, thereby supporting fair artisan evaluation while enhancing the reliability of the review system. (CG = Computer-Generated (fake) review; OR = Original (genuine) review.)

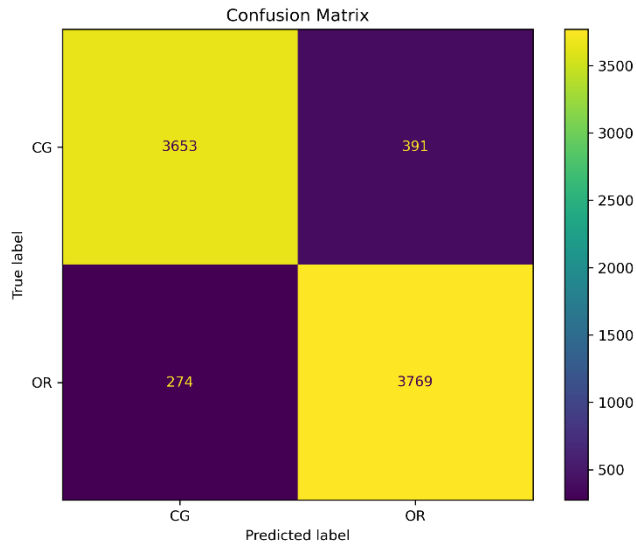


Fig. 3. Confusion matrix of the fraud detection

The discriminative capability of the model is further illustrated by the Receiver Operating Characteristic (ROC) curve shown in Fig. 4. The obtained ROC-AUC score of 97.28% indicates excellent separation between genuine and fraudulent reviews across different decision thresholds, highlighting the robustness of the proposed detection approach.

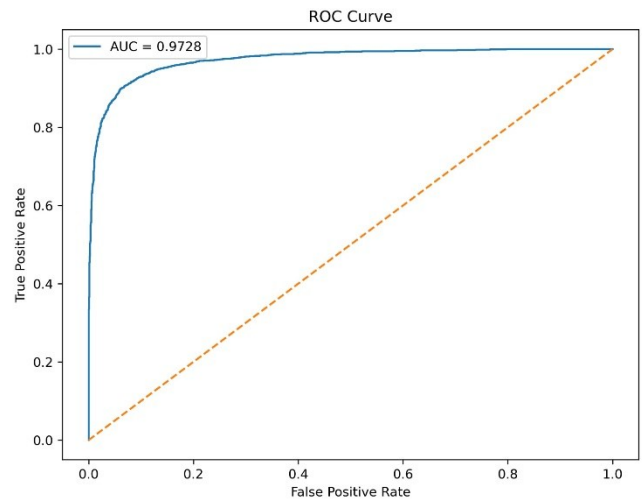


Fig. 4. ROC curve of the fraud detection model

The Precision–Recall curve depicted in Fig. 5 complements the ROC analysis by evaluating the model under varying classification thresholds. Precision remains close to 1.0 over most of the recall range and decreases only at very high recall values, indicating that the proposed model maintains a high level of reliability while detecting the majority of fraudulent reviews. This behaviour is particularly suitable for review moderation systems, where maintaining trust and minimising false accusations are equally important.

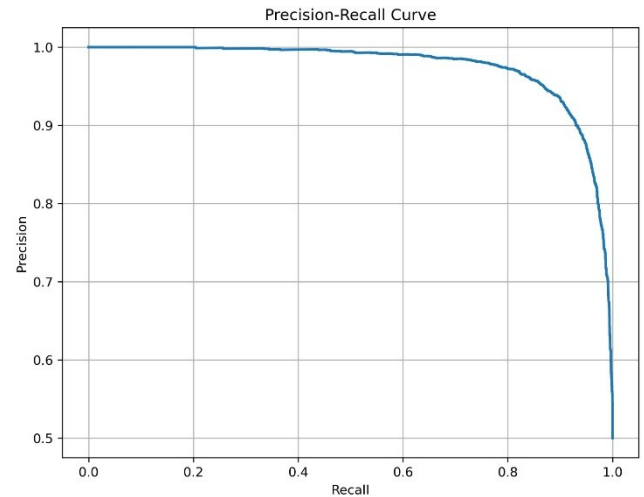


Fig. 5. Precision–Recall curve of the fraud detection model

### 3. Illustrative examples

This section presents the operational workflow and the main user interfaces of the proposed M3ALEM+ platform. The complete service lifecycle, from request creation to payment and review moderation, is illustrated through representative workflows and application screenshots. Each interface highlights a key functionality implemented in the system and demonstrates how clients and artisans interact throughout the service execution process.

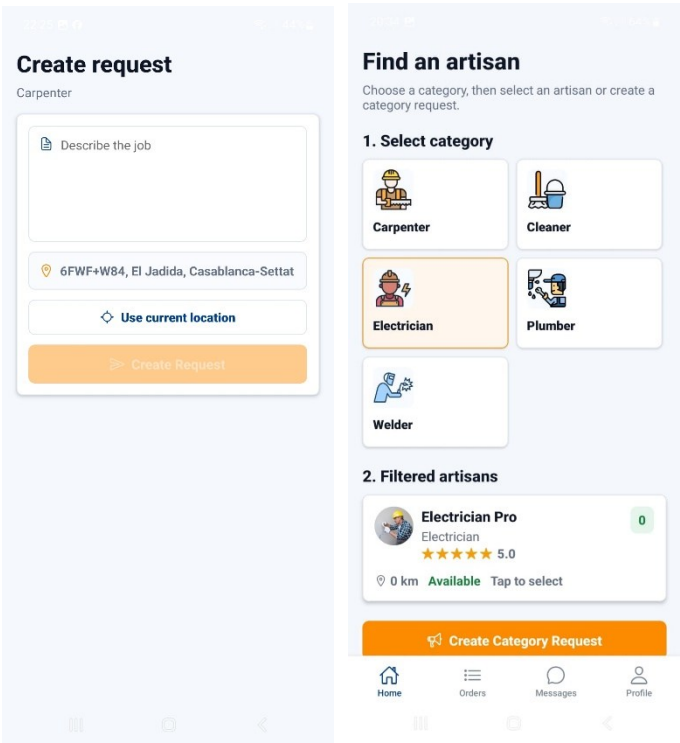


Fig. 6. Workflow of request & matching

Fig.6. illustrates the request matching workflow implemented in M3ALEM+. After a client submits a service request, the backend identifies nearby artisans using GPS coordinates and ranks eligible candidates according to distance, availability, experience, and service rating. The highest-ranked artisans receive real-time notifications and may submit service offers to the client.

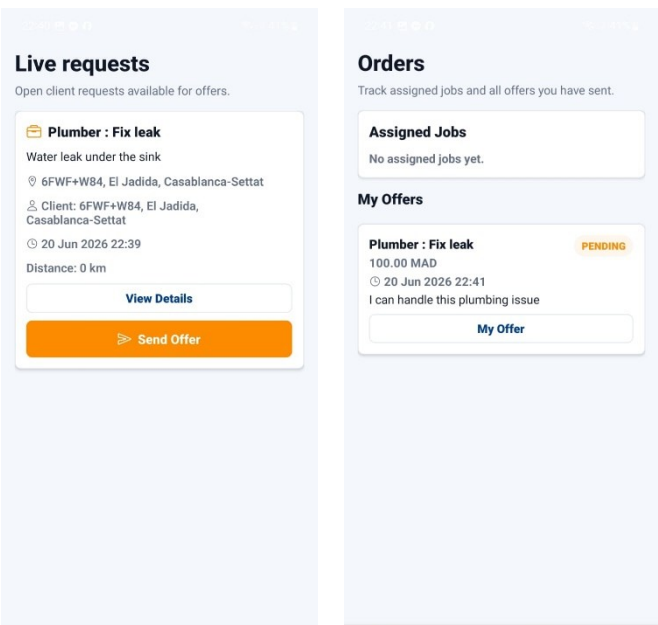


Fig. 7. Workflow of offer

Fig. 7. presents the offer management interface, where clients can review the received proposals and either accept or reject them. Once an offer is accepted, the corresponding artisan is assigned to the request, and both users are notified to initiate the service execution process.

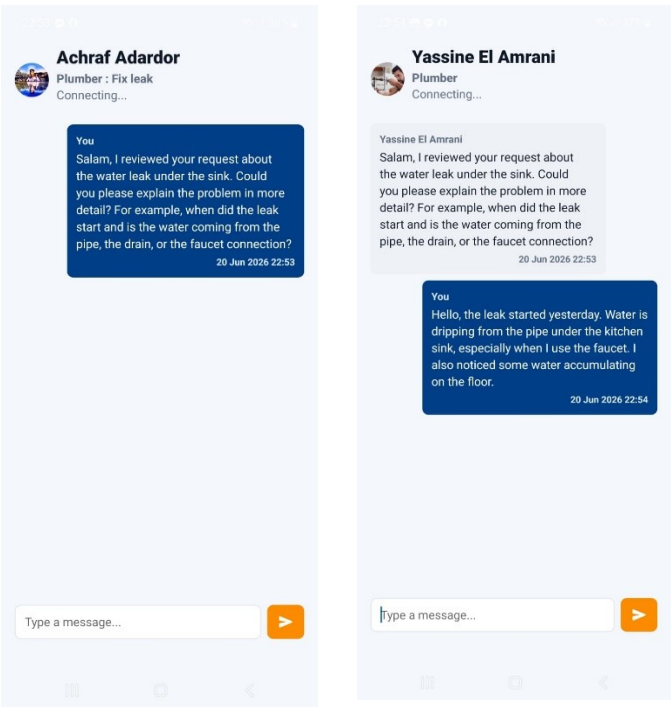


Fig. 8. Chat between client and artisan



Fig.8 shows the integrated real-time messaging interface that enables direct communication between clients and artisans after service assignment. The chat module supports instant message exchange and service status coordination throughout the intervention.

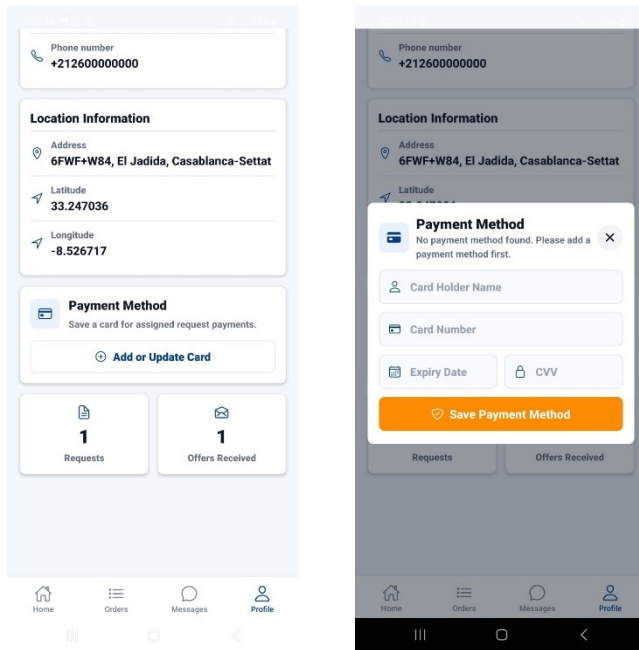


Fig. 9. Payment interface

Fig.9 illustrates the payment interface displayed after service completion. The platform integrates a secure payment workflow, allowing clients to confirm the transaction before submitting a rating and review for the completed service.

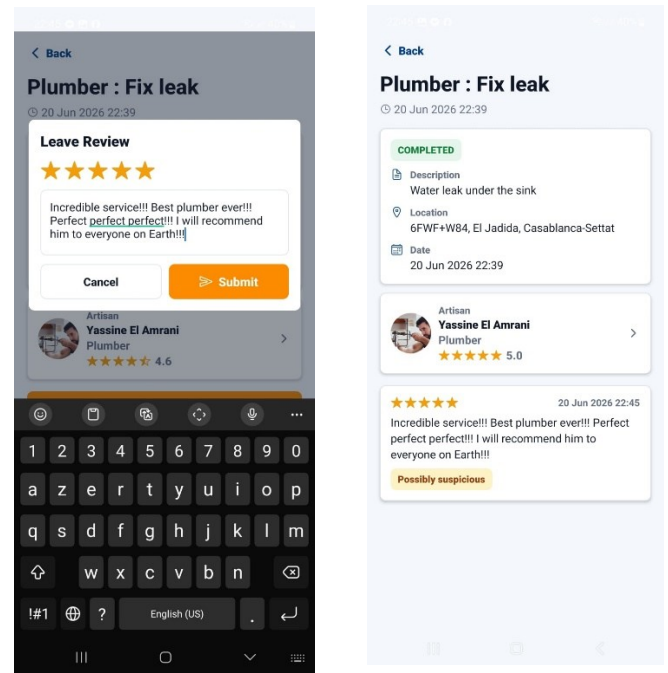


Fig. 10. Fraud detection interface

Fig.10 presents the fraud detection module integrated into the platform. Suspicious reviews identified by the Isolation Forest model are automatically flagged for moderation, preventing manipulated ratings from affecting artisan reputation until manual verification is completed.

#### 4. Comparison with existing platforms

The Moroccan digital services landscape includes several platforms connecting clients with artisans, among which Maalam.ma represents one of the most established solutions. While Maalam.ma laid the foundation for digitising client–artisan interactions in Morocco, it remains limited by a static and manually driven workflow, where service requests are processed asynchronously with response delays ranging from hours to days, communication is handled externally via phone calls, artisan listings are presented without quality-based ranking, no automated fraud detection mechanisms are implemented, and payments are conducted offline without transaction security or traceability.

In contrast, M3ALEM+ is designed to overcome these limitations by introducing a real-time, intelligent service model that enables instant dispatch of nearby artisans, integrated in-app communication, algorithmic ranking based on multiple quality indicators, AI-driven fraud detection, and secure escrow-style payment processing, providing an end-to-end digital service experience.

| Feature              | Maalam.ma                         | M3ALEM+                                             |
|----------------------|-----------------------------------|-----------------------------------------------------|
| Real-time matching   | Not available (manual quotes)     | Instant dispatch with cascading offers              |
| In-app communication | External phone/messaging required | Integrated WebSocket chat                           |
| Artisan ranking      | Chronological listing             | Composite quality scoring + collaborative filtering |
| Fraud detection      | None                              | Isolation Forest anomaly detection                  |
| Payment processing   | Cash only (external)              | Escrow-style secure payment                         |
| Mobile application   | Web-only                          | React Native (iOS + Android)                        |
| Geolocation support  | City-level filtering              | GPS-based proximity queries                         |
| AI recommendation    | None                              | Hybrid collaborative + content-based filtering      |

Table 4. Comparison between Maalam.ma and M3ALEM+

Beyond the functional comparison, M3ALEM+ introduces a fundamentally different operational paradigm. Instead of acting as a static directory of artisans, the platform continuously coordinates request allocation, communication, payment, and reputation management through intelligent services. This integrated workflow improves responsiveness, user experience, and platform trustworthiness while reducing manual intervention.

## 5. Reusability and impact

The proposed M3ALEM+ platform was designed following modular, scalable, and extensible software engineering principles to facilitate long-term maintenance, future development, and research reproducibility. The overall architecture separates the mobile application, backend services, database, and artificial intelligence components into independent layers, thereby reducing coupling between system modules and simplifying future upgrades. Such an architecture enables developers to extend individual components without affecting the remaining parts of the platform, making the system suitable for continuous evolution as new technologies and business requirements emerge.

A major contribution of the proposed architecture is the complete decoupling of the artificial intelligence module from the core application through a dedicated microservice exposed via a versioned REST API. Recommendation and fraud detection models can therefore be independently trained, evaluated, deployed, replaced, or upgraded without interrupting platform operation. This design also enables experimentation with alternative machine learning algorithms, deep learning models, or hybrid recommendation techniques while preserving compatibility with the existing application. Furthermore, model retraining may be executed periodically using newly collected platform data or triggered manually by administrators, allowing the recommendation engine and fraud detection models to

continuously improve over time without requiring changes to the mobile application or backend business logic.

The platform additionally employs a database-driven service taxonomy rather than a hard-coded service catalogue. As a result, new artisan categories, professions, or service domains can be introduced simply by updating the database, eliminating the need for software modifications or recompilation. This flexibility enables the proposed architecture to be adapted to numerous application scenarios beyond artisan services, including healthcare assistance, elderly care, home maintenance, tutoring, transportation, logistics, technical support, and other location-based professional services. Consequently, M3ALEM+ should be viewed as a generic location-based service platform rather than a domain-specific application.

To improve portability and reproducibility, all software components are containerised using Docker Compose. The complete deployment environment—including the backend server, database, AI microservice, and supporting services—can therefore be reproduced consistently across development, testing, academic, and production environments. This deployment strategy simplifies software installation, facilitates collaborative development, and allows other researchers to reproduce experiments under identical execution conditions. In addition, environment-based configuration management enhances deployment security and portability by eliminating hard-coded credentials and platform-specific configuration parameters.

From a software engineering perspective, the proposed architecture also supports horizontal scalability. Since the AI module operates independently from the main backend, computationally intensive tasks such as recommendation generation and fraud detection can be executed on dedicated servers without affecting user-facing services. This separation facilitates future deployment in cloud environments and enables resource allocation according to computational demand, thereby improving both responsiveness and scalability as the number of users increases.

Beyond its technical contributions, M3ALEM+ has significant socio-economic implications. In many developing countries, artisan services remain largely informal, making it difficult for clients to identify qualified professionals while limiting artisans' visibility and access to new customers. By providing a secure digital marketplace that integrates geolocation, reputation management, verified reviews, and secure payment mechanisms, the platform contributes to the progressive digitalisation of local service ecosystems. Artisans benefit from increased visibility, transparent reputation building, and improved access to employment opportunities, whereas clients benefit from reliable service recommendations, transparent pricing, secure transactions, and improved trust throughout the service lifecycle.

The integration of an AI-based fraud detection module further strengthens platform reliability by automatically identifying suspicious behavioural patterns and potentially manipulated reviews before they influence artisan rankings. This capability reduces the risk of reputation manipulation, promotes fair competition among service providers, and increases overall confidence in recommendation results. Unlike conventional review systems that rely exclusively on manual moderation, the proposed approach combines automated detection with



administrative verification to provide an additional layer of platform security.

From a research perspective, M3ALEM+ constitutes a reusable experimental framework for studies involving intelligent mobile marketplaces, location-based service recommendation, geospatial information systems, and trustworthy AI applications. Its modular architecture enables researchers to integrate alternative recommendation algorithms, explainable AI techniques, reinforcement learning strategies, or advanced fraud detection approaches while preserving the overall software infrastructure. Consequently, the platform can serve not only as a production-ready application but also as an experimental testbed for future research on intelligent service ecosystems.

Although the current implementation demonstrates the successful integration of geolocation, real-time communication, secure payment processing, recommendation systems, and AI-assisted fraud detection within a unified mobile platform, several research directions remain open. Future versions will incorporate multilingual interfaces supporting Arabic, French, and Tamazight to improve regional accessibility, real-time artisan tracking throughout service execution, and advanced natural language processing techniques for sentiment analysis and review authenticity verification. Additional developments include personalised recommendation models based on user behaviour, predictive demand forecasting using historical service requests, dynamic pricing assistance, premium visibility mechanisms for professional artisans, integration with external mobility and navigation services for improved arrival-time estimation, and the exploration of explainable artificial intelligence techniques to increase transparency in recommendation and fraud detection decisions.

## 6. Conclusion

This paper presented **M3ALEM+**, an intelligent and modular mobile marketplace designed to modernise on-demand artisan services through the integration of geolocation technologies, real-time communication, secure digital payments, personalised recommendation, and artificial intelligence-based fraud detection. Unlike conventional service marketplaces that primarily facilitate client–artisan connections, the proposed platform combines modern software engineering practices with machine learning techniques to provide a secure, scalable, and trustworthy service ecosystem.

The proposed architecture demonstrates how mobile technologies, cloud-based microservices, and lightweight artificial intelligence models can be effectively integrated into a unified software solution while preserving modularity, extensibility, and deployment flexibility. The separation of the AI components into an independent microservice enables recommendation and fraud detection models to evolve independently from the core application, facilitating future maintenance, experimentation, and continuous model improvement without affecting platform availability.

Experimental evaluation demonstrates that the integrated fraud detection module successfully identifies suspicious reviews while maintaining high classification performance, thereby

increasing the reliability of the reputation system and improving the quality of artisan recommendations. Combined with geolocation-based matching, real-time messaging, and secure payment management, these capabilities contribute to a more transparent, efficient, and trustworthy digital service marketplace.

Overall, M3ALEM+ represents both a practical solution for the digital transformation of artisan service ecosystems and a reusable reference architecture for future research on intelligent mobile marketplaces, location-based services, trustworthy recommendation systems, and AI-enabled service platforms. Future research will focus on multilingual support, real-time artisan tracking, explainable and personalised recommendation models, advanced natural language processing for sentiment analysis and review authenticity verification, predictive demand forecasting, and further optimisation of fraud detection techniques to continuously enhance platform intelligence and user experience.

## CRedit authorship contribution statement

Achraf: Conceptualization, Methodology, Software, Writing – original draft, Visualization.

Bouchra: Conceptualization, Methodology, Software, Validation, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

## Acknowledgements

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## Data availability

The source code associated with this work is openly available at:

<https://github.com/achraf-bouchra/m3alem-plus>

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